Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25.parquet

SimpleRNN Multi-Step Training Report

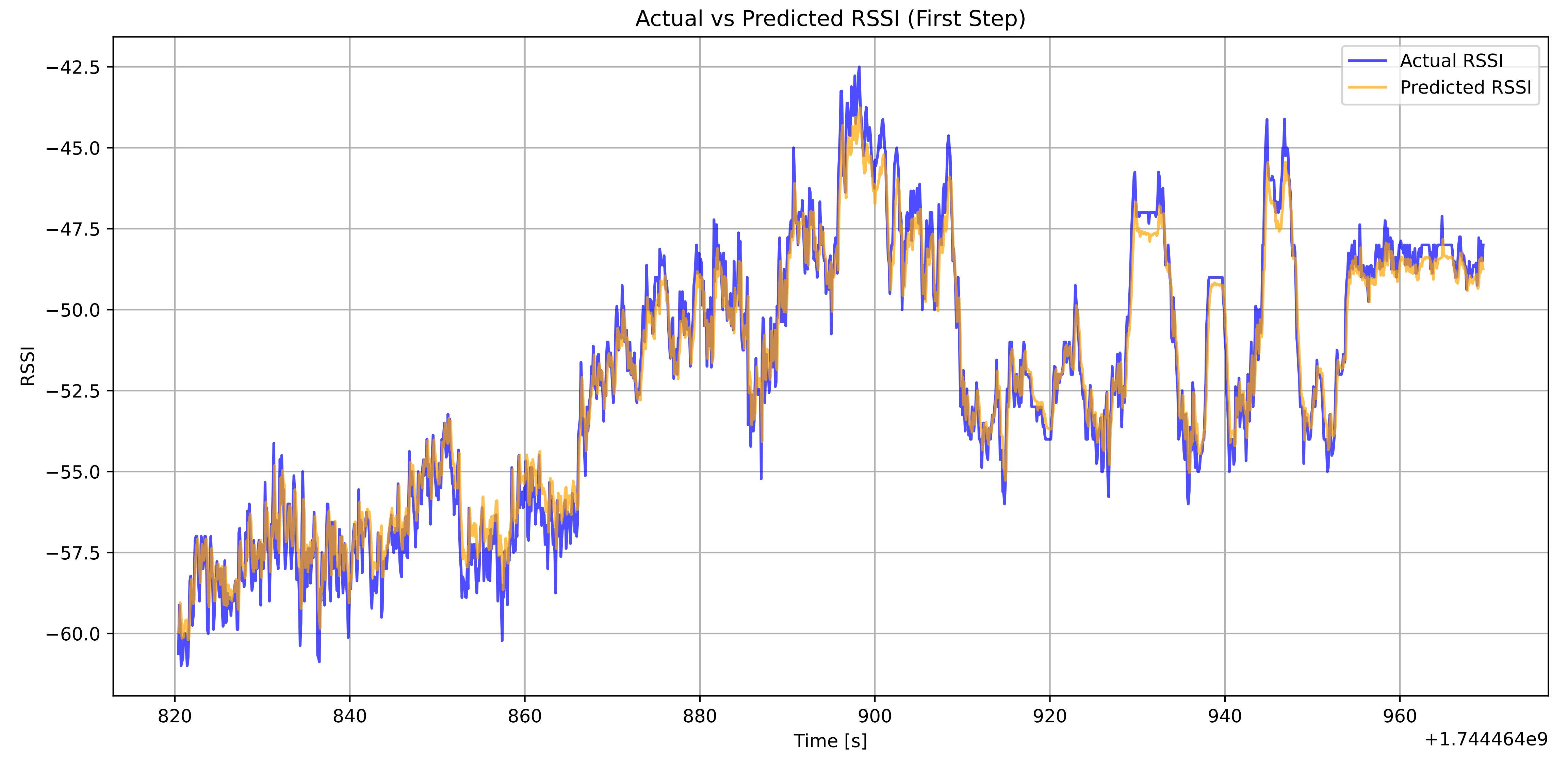
Report Generated: 2025-05-07 09:35:51

Training Start: 2025-05-07_09-24-04
Training End: 2025-05-07_09-27-32
Training Duration: 3.47 minutes

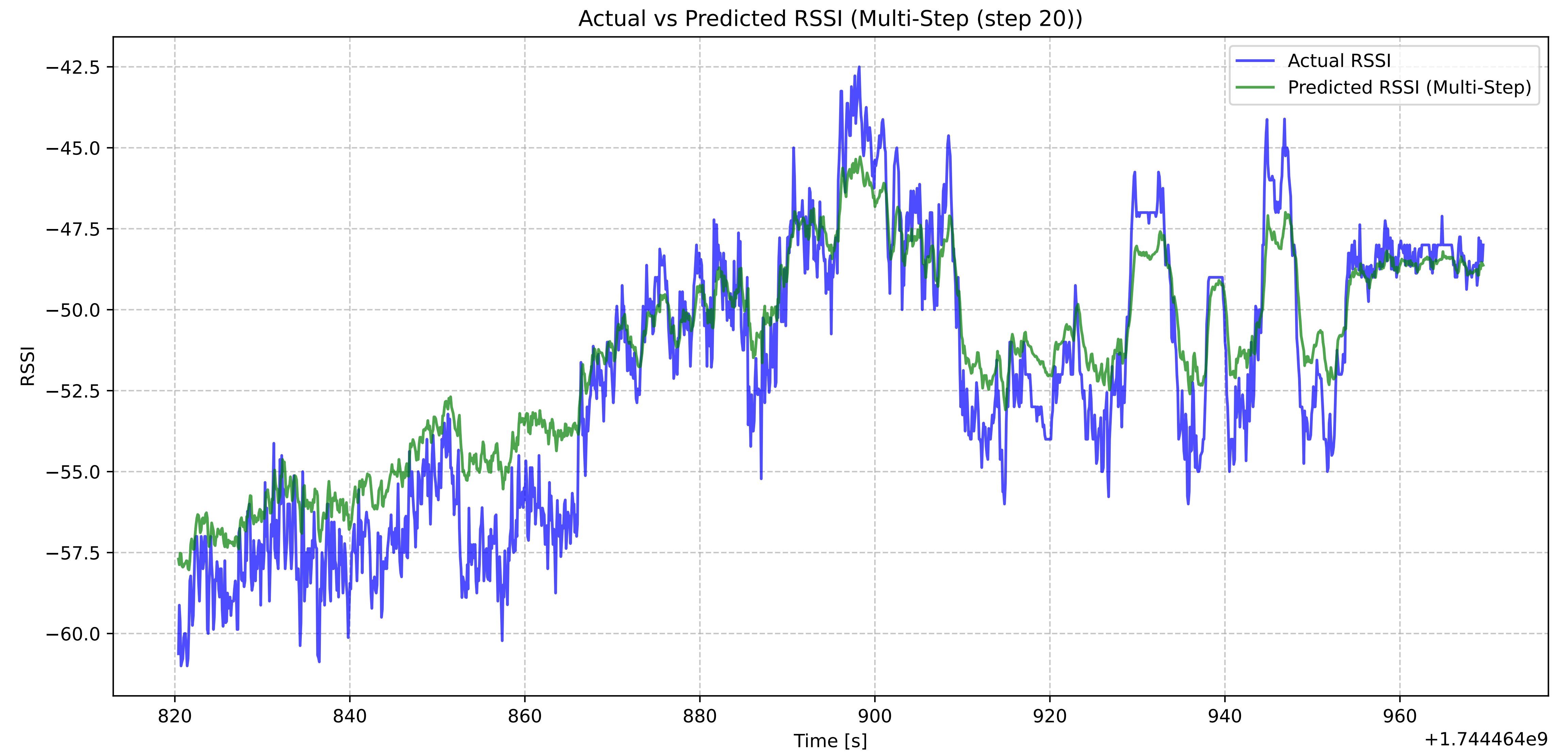
Best Hyperparameters:
Number of Layers: 2
Activation: tanh
Optimizer: Nadam
Dropout Rate: 0.20

MAE: 1.4161
MSE: 3.4725
RMSE: 1.8635
R2 Score: 0.7962

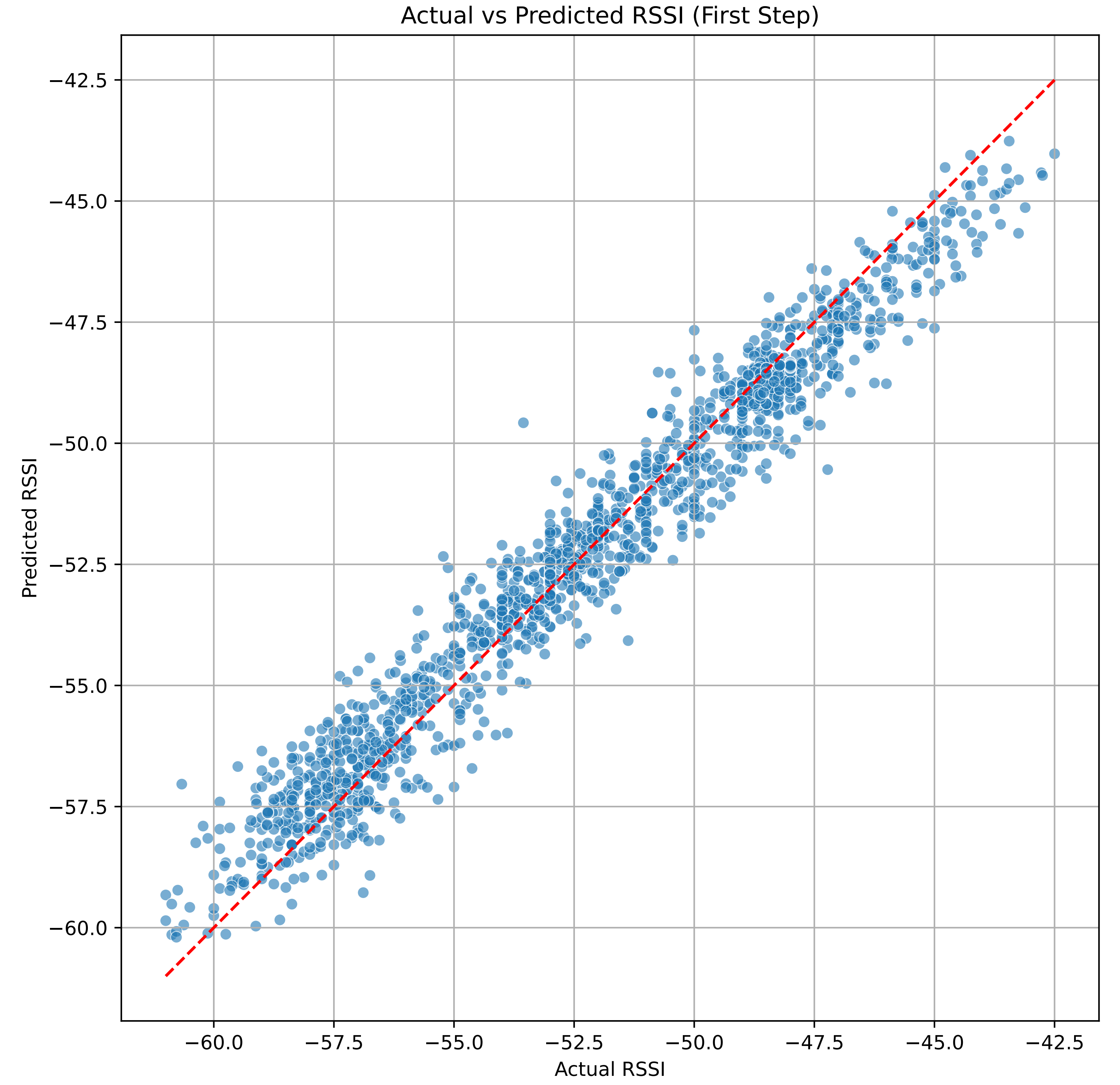
Multi-Step Forecast:
Forecast Steps: 20
MAE: 0.0261
MSE: 0.0011
RMSE: 0.0331
R2 Score: 0.9534



Actual vs Predicted RSSI (First Step)
Actual RSSI
Predicted RSSI
RSSI
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
820
840
860
880
900
920
940
960
Time [s]
+1.744464e9

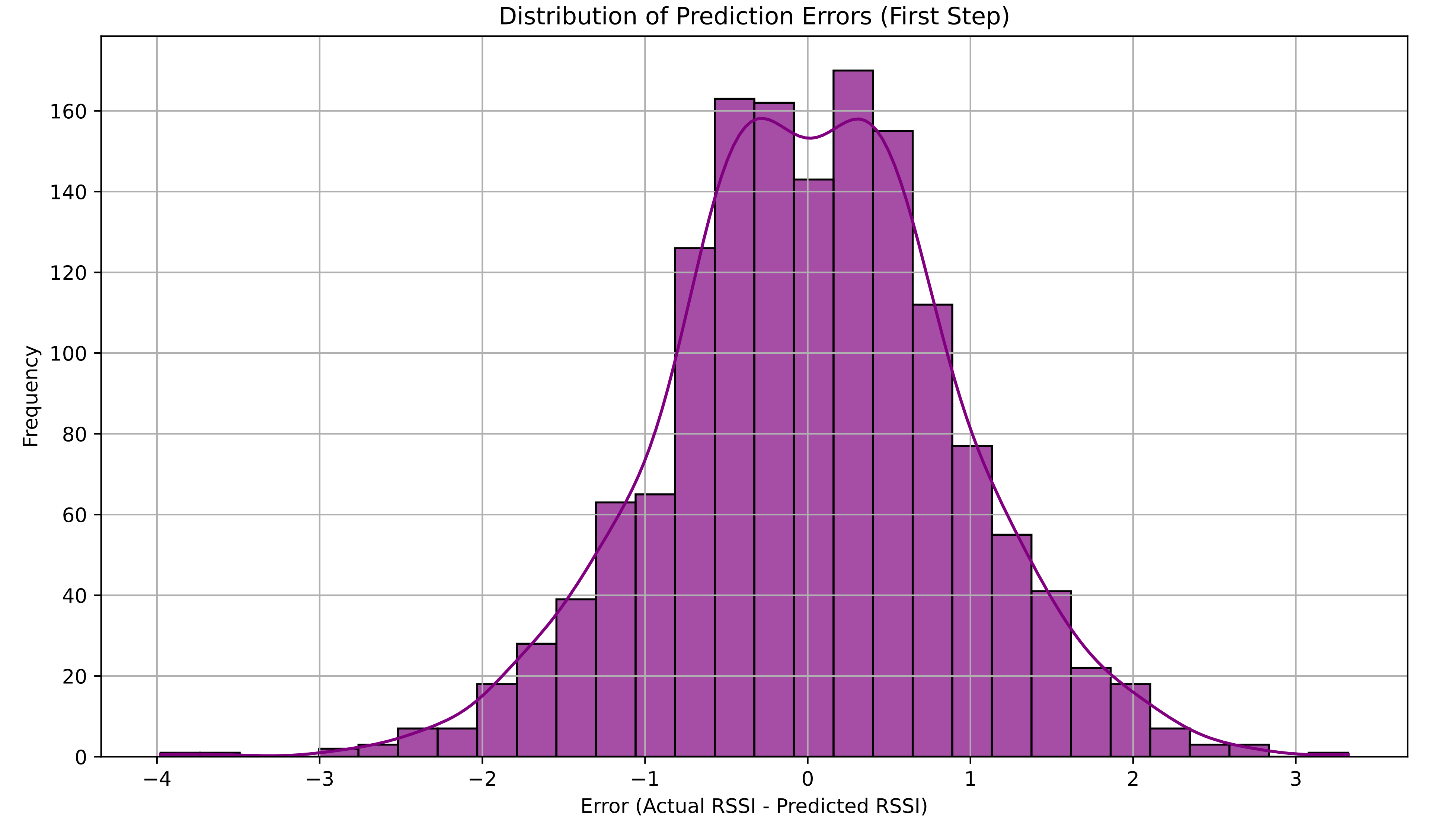


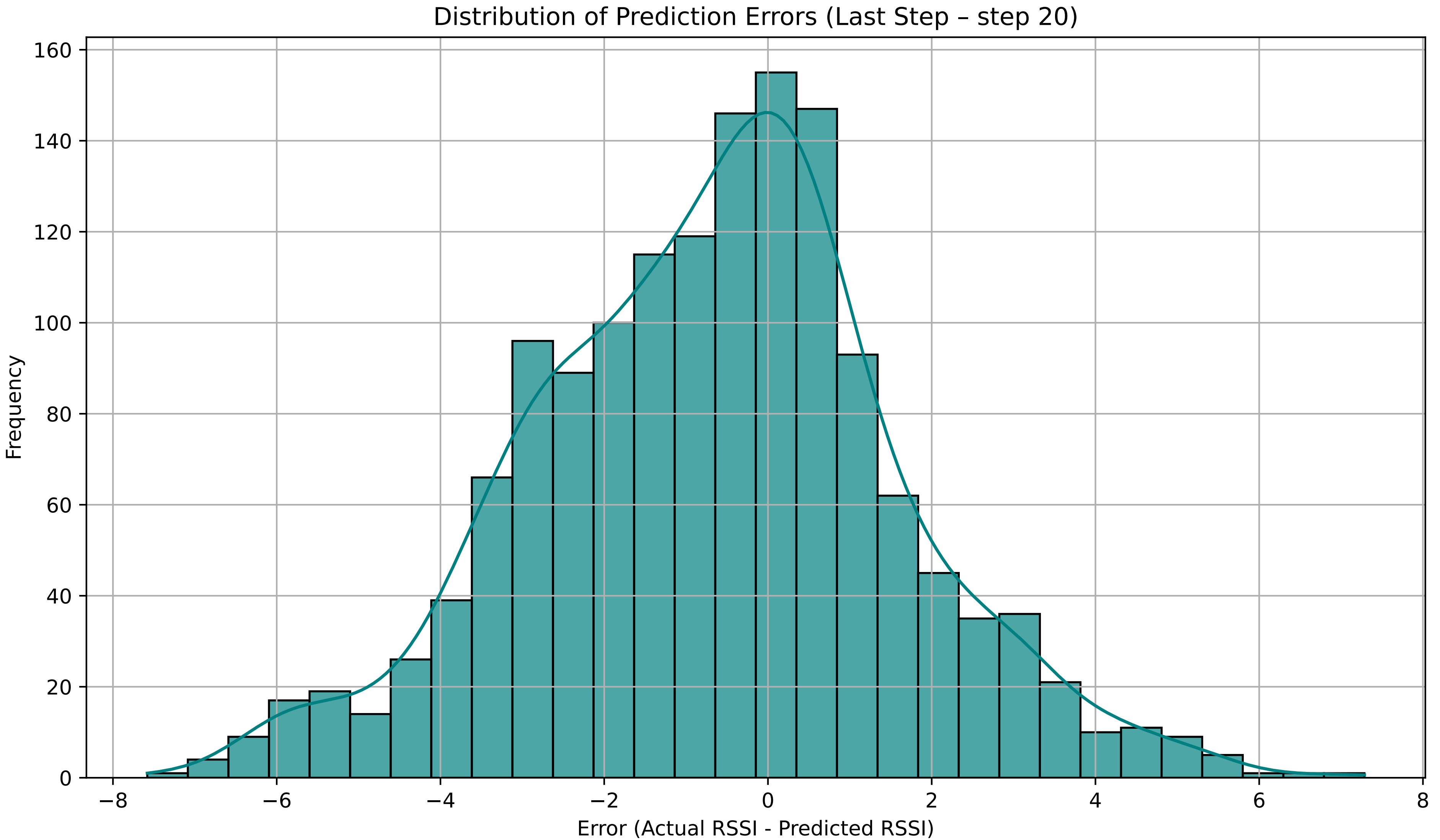
Actual vs Predicted RSSI (Multi-Step (step 20))
Actual RSSI
Predicted RSSI (Multi-Step)
RSSI
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
820
840
860
880
900
920
940
960
Time [s]
+1.744464e9

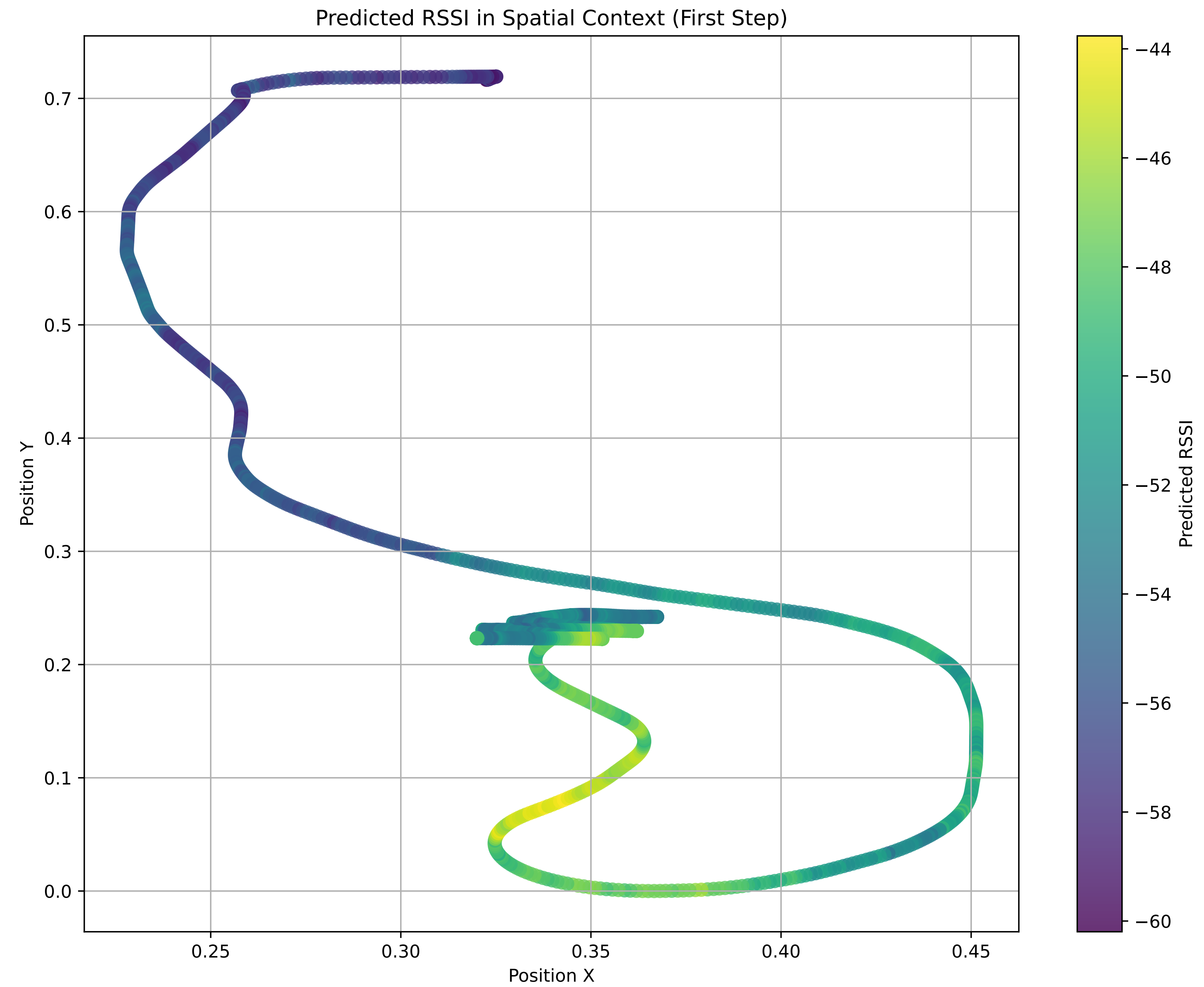


Actual vs Predicted RSSI (First Step)
Predicted RSSI
Actual RSSI
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
−60.0
−57.5
−55.0
−52.5
−50.0
−47.5
−45.0
−42.5

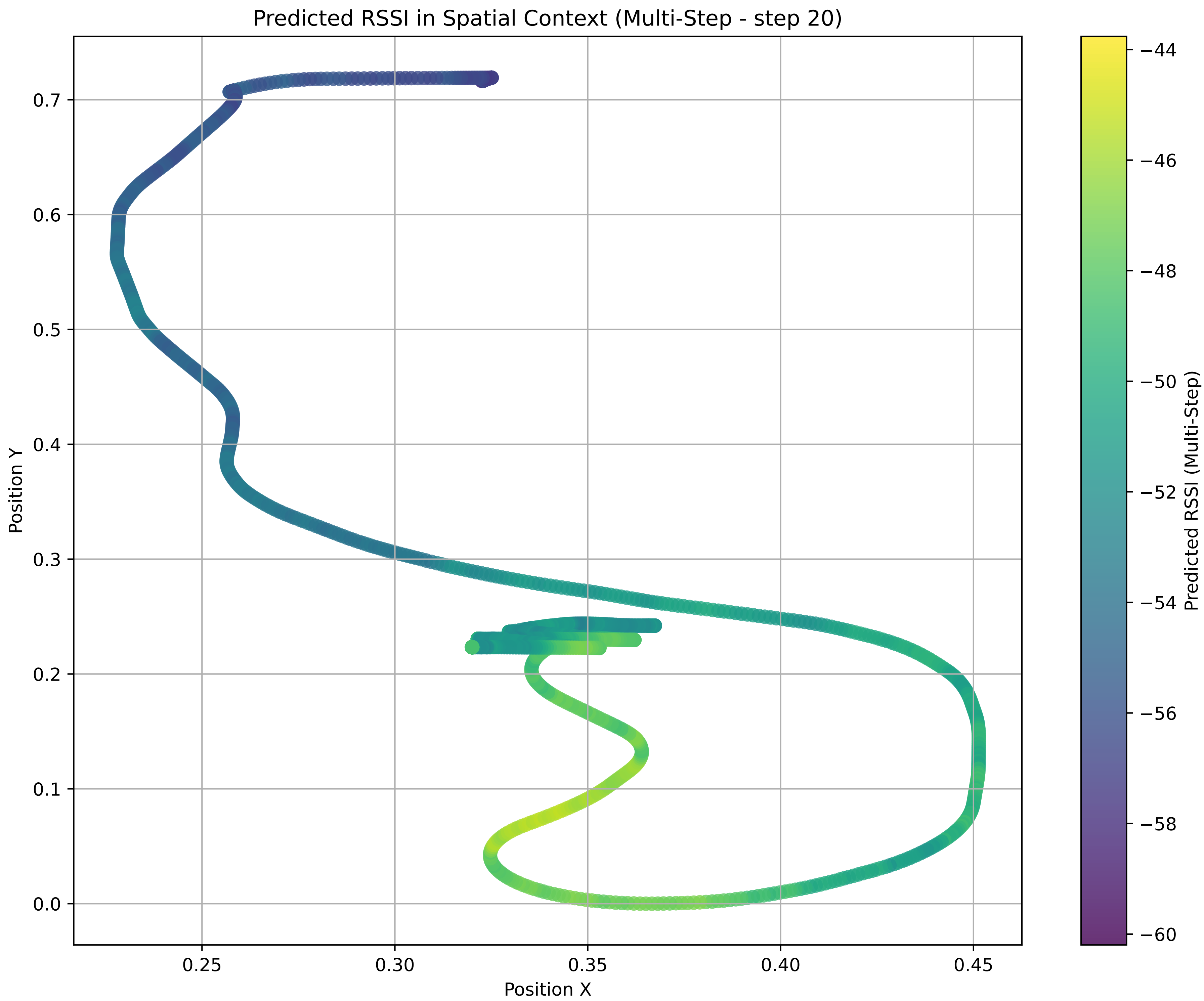
Actual vs Predicted RSSI (Last Step – step 20)
Predicted RSSI
Actual RSSI
−42.5
−45.0
−47.5
−50.0
−52.5
−55.0
−57.5
−60.0
−60.0
−57.5
−55.0
−52.5
−50.0
−47.5
−45.0
−42.5

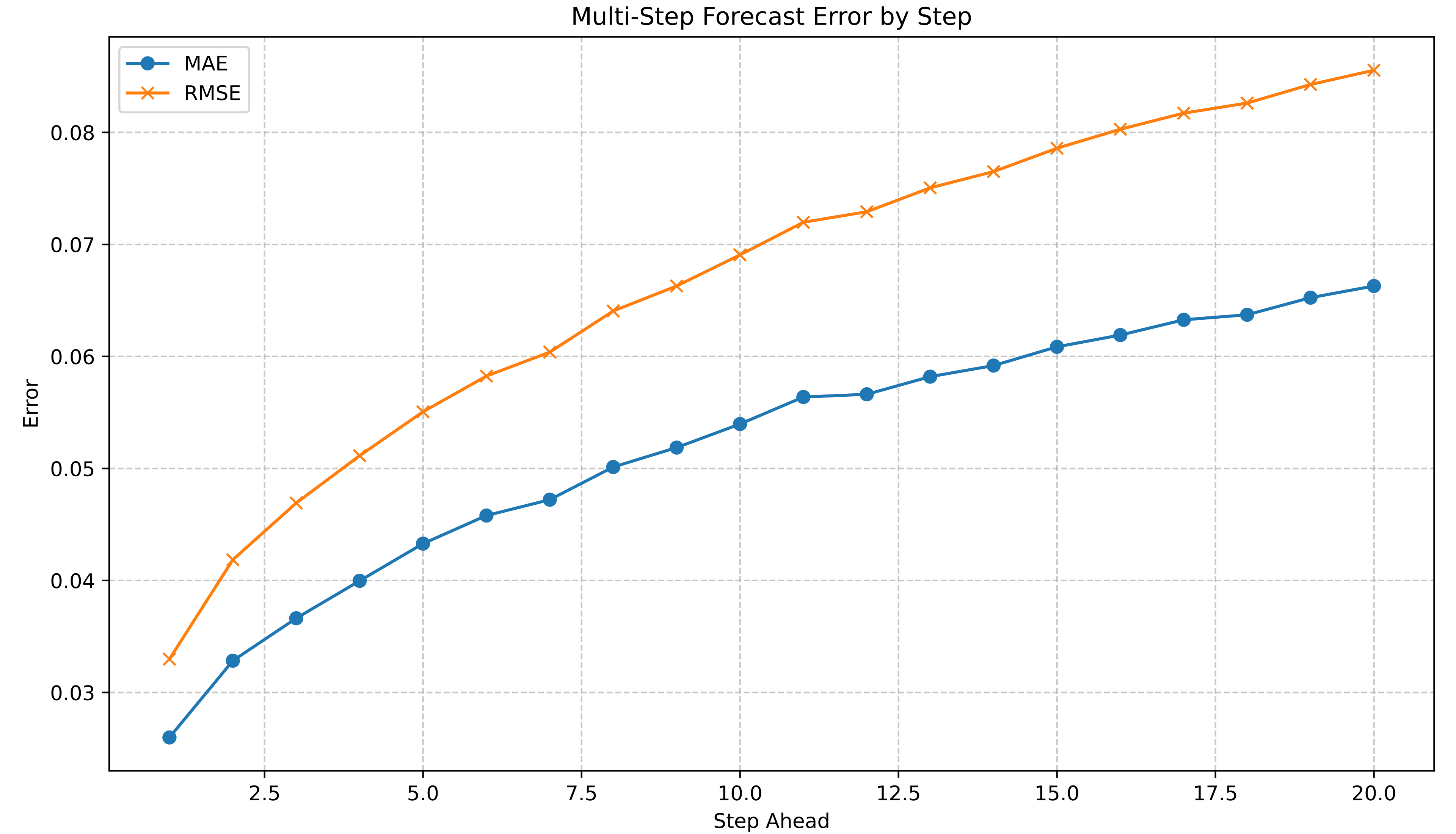

Distribution of Prediction Errors (First Step)
Frequency
0
20
40
60
80
100
120
140
160
−4
−3
−2
−1
0
1
2
3
Error (Actual RSSI - Predicted RSSI)


Distribution of Prediction Errors (Last Step – step 20)
Frequency
0
20
40
60
80
100
120
140
160
−8
−6
−4
−2
0
2
4
6
8
Error (Actual RSSI - Predicted RSSI)



Predicted RSSI in Spatial Context (First Step)
Position Y
Position X
Predicted RSSI
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.25
0.30
0.35
0.40
0.45
−44
−46
−48
−50
−52
−54
−56
−58
−60


Predicted RSSI in Spatial Context (Multi-Step - step 20)
Position Y
Position X
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.25
0.30
0.35
0.40
0.45
Predicted RSSI (Multi-Step)
−44
−46
−48
−50
−52
−54
−56
−58
−60


Multi-Step Forecast Error by Step
MAE
RMSE
Error
0.03
0.04
0.05
0.06
0.07
0.08
Step Ahead
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

```
Resumo Automático
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Dataset:        rssi_odom_dataset_12apr25.parquet
Modelo:         SimpleRNN
Forecast steps: 20

MAE / RMSE (dB)
    • 1º passo : 0.03 / 0.03
    • Mediana  : 0.06 / 0.07
    • 20º passo : 0.07 / 0.09

Crescimento do erro: 155.1%   →   Degrada Rápido (> 50 %)

3 piores passos (MAE): 20 (0.07), 19 (0.07), 18 (0.06)

Recomendação:
Considere aumentar o histórico de entrada ou usar um modelo mais
profundo para melhorar multi-step.
```